

Alireza Jaberirad

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RESEARCH INTERESTS

Hardware–software co-design for efficient ML inference • Noise-tolerant computing • In-memory computing

EDUCATION

University of Massachusetts Amherst Sep 2023 – May 2028 (Expected)
PhD in Computer Engineering | Focus: AI Hardware Acceleration | GPA: 3.68/4.0 Amherst, MA

University of Tehran Sep 2017 – Jun 2022
Dual B.Sc. in Electrical Engineering and Computer Science Tehran, Iran

EE focus: Digital System Design, GPA: 3.70/4.0, Ranked 11/135 | *CS focus:* AI, GPA: 3.71/4.0, Ranked 3/40

Selected Coursework: Neuromorphic Engineering, VLSI Design, FPGA-Based Embedded System Design, Machine Learning, Artificial Intelligence, Algorithms

TECHNICAL SKILLS

Languages: Python, C, C++, Verilog, MATLAB, SystemC
ML / AI: PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, NLP, HDC, Event-based Learning
FPGA & Embedded: Intel/Altera and Xilinx FPGAs, Quartus, Qsys, Nios II, Avalon, ModelSim, STM32
IC & Hardware Design: Cadence Virtuoso, Cadence Innovus, HSPICE, LTspice, Altium Designer
Tools & Workflow: Git, Linux, Tetramem SDK

RESEARCH EXPERIENCE

Analog Logic Circuits Dec 2024 – Present
Manuscript in preparation for IEEE Transactions on Circuits and Systems

- Built a simulator that maps digital logic netlists onto analog memristor hardware and models latency.
- Achieved **2.8 – 43× lower latency** and **7.4 – 113× lower energy** than an equivalent digital ASIC.
- Quantified the precision–accuracy trade-off under device variation across 19 benchmark circuits.
- Skills: Hardware performance modeling, design-space exploration, numerical simulation, Python, NumPy

Hyperdimensional Computing – Hardware/Algorithm Co-Design May 2024 – May 2025
Published at MLNCP Workshop @ NeurIPS 2024 | Journal extension published at Nature Communications

- Co-designed a quantized HDC inference algorithm using device stochasticity, cutting model memory **90%**.
- Achieved best-reported accuracy on 21-language classification: **99.35%** software vs. **95.24%** on-chip.
- Held **98.12%** accuracy on cancer classification under analog device noise; extended to NLP and vision tasks.
- Skills: Quantization-aware algorithm design, inference optimization, PyTorch, Tetramem SDK, Linux

3D Memristive Arrays for Single-Step Matrix Operations Sep 2023 – Jan 2026
Journal paper published at Advanced Intelligent Systems

- Designed a retina-inspired 3D architecture for one-shot fully parallel matrix multiplication and convolution.
- Achieved **92.96%** and **89.31%** precision on edge-detection tasks on CMOS and nanoscale prototypes, respectively.
- Delivered **113×** lower energy and orders-of-magnitude lower latency than the CMOS digital baseline.
- Skills: Computer architecture, C++, MATLAB, microcontroller programming

FPGA-Based Audio Processing Accelerator Sep 2021 – Jan 2022
Embedded Systems Laboratory, University of Tehran Tehran, Iran

- Accelerated a 64-tap FIR audio filter with a **Nios II custom instruction**: 50 minutes down to **2–3 seconds**.
- Mapped the filter fully parallel and fully serial: **16,993** logic elements and 70 DSPs versus **2,081** and 4.
- Designed a parameterized fixed-point FIR datapath in Verilog, pipelined to reach **197 MHz** on a Cyclone II.
- Skills: Verilog, Quartus, Qsys, Nios II, Avalon burst DMA, ModelSim, fixed-point arithmetic, C

PUBLICATIONS

Published and Accepted Papers (* co-first author)

1. **A. Jaber Rad**, et al. “Single-Step Analogue In-Memory Matrix Computation in Three-Dimensional Circuits.” *Advanced Intelligent Systems*, 2026. [DOI]
2. Y. Huang*, **A. Jaber Rad***, et al. “Hyperdimensional In-Memory Computing with Analogue Memristive Crossbar Arrays.” *Nature Communications*, 2026. [DOI]
3. Y. Huang, **A. Jaber Rad**, et al. “Hardware–Algorithm Co-Design for Hyperdimensional Computing Based on Memristive System-on-Chip.” *MLNCP Workshop @ NeurIPS*, 2024. [OpenReview, arXiv]
4. W. Zhao, Y. Huang, A. Tewari, **A. Jaber Rad**, et al. “Event-based Neuromorphic Sensing System with Flexible Haptic Sensors and a Memristive System on a Chip.” *Nature Sensors*, 2026. [DOI]
5. C. He, Y. Huang, A. Korkmaz, **A. Jaber Rad**, et al. “Cognitive Radio Receiver with Analogue In-Memory Computing.” *Science Advances* (accepted).

Manuscripts Under Review

6. Y. Ling, Z. Xiong, Y. Huang, **A. Jaber Rad**, et al. “Tunable Diffusive Memristor Arrays for Neuromorphic Temporal Computing.” *Nature Materials*.

WORK EXPERIENCE & SERVICE

Peer Reviewer

2025 – 2026

The Institute of Electrical and Electronics Engineers (IEEE)

- Reviewed 3 manuscripts for *IEEE Transactions on Electron Devices*.
- Reviewed 2 manuscripts for *IEEE International Symposium on Circuits and Systems (ISCAS)* 2026.

Summer Engineering Institute (SENGI) Supervisor

Jul 2024

University of Massachusetts Amherst

Amherst, MA

- Led a cohort of K–12 students in building a UI for a hyperdimensional-computing software simulation.
- Taught fundamentals of neural networks and hyperdimensional computing to a non-specialist audience.
- Coached students on clean code and Python tooling: NumPy, Pandas, Tkinter, Matplotlib.
- Mentored team-based development workflows in VS Code Live Share, simulating an industry collaboration setting.

Automotive Embedded Software Intern

Jun 2021 – Sep 2021

Irankhodro Powertrain Co. (IPCO)

Tehran, Iran

- Developed embedded C/C++ firmware on STM32 for an automotive ECU (ignition, immobilizer, keyless entry).
- Validated ECU functionality through MCU-based prototyping with simulated input signals.

PRESENTATIONS

1. Talk, *UMass Graduate Research Symposium*, 2026. “Seeing Like the Eye, Thinking Like the Brain for Faster and Greener AI.”
2. Poster, *UMass ECE Graduate Research Poster Session*, 2025. “Hardware–Algorithm Co-Design for Hyperdimensional Computing Based on Memristive System-on-Chip.”
3. Talk, *UMass ECE 3MT Competition*, 2025. “A New Era of AI: Efficient Software + Brain-Inspired Hardware.”
4. Poster, *MLNCP Workshop @ NeurIPS*, Vancouver, 2024. “Hardware–Algorithm Co-Design for Hyperdimensional Computing Based on Memristive System-on-Chip.”

HONORS & AWARDS

Lightning-Talk Presenter , UMass Graduate Research Symposium (1 of 10 selected).	2026
David H. Navon Scholarship Award for outstanding research contribution in microelectronics.	2025
Second Place , ECE Graduate Research Poster Session, UMass Amherst.	2025
Best Undergraduate Project Award for designing an ASIC neural-network accelerator, University of Tehran.	2022
Exceptional-Talent M.Sc. Admission Award for Sharif University of Technology and University of Tehran.	2022
FOE Award for ranking in the top 10% of both EE and CS cohorts, University of Tehran.	2019
Tuition-Free Second Major , awarded to 103 of 32,000 qualifying University of Tehran students.	2018
Rank 37 of 150,000 , Iranian National University Entrance Examination.	2017
Semi-Finalist , Iranian National Mathematics Olympiad and National Computer Olympiad.	2015–2016